Recall F field, $n \in \mathbb{Z}$, then

$$\begin{aligned} & \{F[X]\text{-modules}, \dim_F = n\}/\simeq \\ \longleftrightarrow\ & \{(V,T) : \dim V = n, T \in \mathrm{End}(V)\}/\approx \\ \longleftrightarrow\ & M_{n\times n}(F)/\text{conjugation} \end{aligned}$$

$f = c_0 + \ldots + c_{n-1}X^{n-1} + X^n \in F[X]$, then $F[X]/(f) \longleftrightarrow C_f$ up to $\simeq$ and conjugation, respectively.

Structure Theorem R PID, M finitely generated R-module, then

$$M \simeq R/I_1 \oplus \ldots \oplus R/I_k \oplus E$$

where $R \neq I_1 \supset \ldots \supset I_k \neq \{0\}$ ideals, $k \in \mathbb{Z}_{\geq 0}$, E free module, all unique up to $\simeq$.

$R = F[X]$, can write $I_i = (f_i)$, $f_i \in F[X] \setminus F$ is monic / 首一.

Theorem Fix $n \in \mathbb{Z}_{\geq 1}$. Let $A \in M_{n\times n}(F)$, then $\exists!$ sequence of monic $f_1 \mid \ldots \mid f_k$ in $F[X] \setminus F, k \geq 1$ s.t. $\sum \deg(f_i) = n$ and $A \underset{\text{conj}}{\sim} \mathrm{diag}\left(C_{f_1}, \ldots, C_{f_k}\right)$, called the **rational canonical form** / 有理标准形, $f_1 \mid \ldots \mid f_k$: the invariant factors of A/conj.

Proof Let M be the $F[X]$-module$/\simeq$ corresponding to A/conj. Then

$$M \simeq F[X]/(f_1) \oplus \ldots \oplus F[X]/(f_k) \oplus E$$

$\dim_F M = n, \dim_F F[X] = \infty \Longrightarrow E = \{0\}, \sum \dim_F F[X]/(f_i) = \sum \deg f_i = n.$

Known: $F[X]/(f_i) \longleftrightarrow C_{f_i}$, hence $M \longleftrightarrow \mathrm{diag}\left(C_{f_1}, \ldots, C_{f_k}\right)$ up to $\simeq$ and conjugation, respectively. Uniqueness follows from structure theorem. □

Corollary
- The minimal polynomial $\mathrm{Min}_A = f_k$.
- The characteristic polynomial $\mathrm{Char}_A = \prod f_i$.

Proof

- $\forall h \in F[X], h(A) = 0.$

  Multiplication by X on $M = \dfrac{F[X]}{(f_1)} \oplus \ldots \oplus \dfrac{F[X]}{(f_k)} \longleftrightarrow A$ as a linear map $F^n \to F^n$, so

  $$\begin{aligned} h(A) = 0_{n\times n} &\Longleftrightarrow h(X) \text{ acts as } 0 \text{ on } M \\ &\Longleftrightarrow \forall i, h(X) \text{ acts as } 0 \text{ on } F[X]/(f_i) \\ &\Longleftrightarrow f_i \mid h, \forall i \Longleftrightarrow f_k \mid h. \end{aligned}$$

  Thus $\mathrm{Min}_A = f_k$. □
- Suffices to show $\forall i, \mathrm{Char}_{C_{f_i}} = f_i$: known. □

In particular: $\mathrm{Min}_A \mid \mathrm{Char}_A$, i.e. $\mathrm{Char}_A(A) = 0_{n\times n}$: gives another proof of Cayley-Hamilton Theorem.

Remark Sometimes, rational canonical form are stated using ${}^tC_f$. Since $A \underset{\text{conj}}{\sim} B \Longleftrightarrow {}^tA \underset{\text{conj}}{\sim} {}^tb$, this is equivalent.

Another application of structure theorem $R = \mathbb{Z} \Longrightarrow$ Every finitely generated group A,

$$A \simeq \mathbb{Z}/d_1\mathbb{Z} \oplus \ldots \oplus \mathbb{Z}/d_k\mathbb{Z} \oplus \mathbb{Z}^r$$

for unique $d_1 \mid \dots \mid d_k, k \geq 0, \forall d_i \geq 1$ and $r \geq 0$.

Next:

- Variants of structure theorem,
- Application to rational canonical form $\to$ Jordan canonical forms,
- Proof of structure theorem: existence & uniqueness, + computational recipe.

Definition R commutative ring, $I \subset R$ ideal, M R-module. Let

$$M[I] := \{x \in M \mid \forall a \in I, ax = 0\}$$

then $M[I]$ is a submodule of M. When $I = (h)$, we write $M[h] \Longrightarrow M[I] = \bigcap_{h \in I} M[h]$. If R integral domain, $x \in M, x \in M[h]$ for some $h \in R \setminus \{0\}$, we say x is a **torsion element** / 挠元 of M, otherwise we say x is torsion-free.

Example R integral domain, then there are no torsion elements $\neq 0$ in free R-modules. ($\because$ reduce to $M = R$.)

For integral domain R and R-module M,

$$M_{\text{tors}} := \{x \in M : \text{torsion element}\}$$

Claim: M_{tors} is a submodule of M.

- $0 \in M_{\text{tors}}$,
- If $x, y \in M_{\text{tors}}$, say $rx = sy = 0$, then $rs(x+y) = 0 \Longrightarrow x + y \in M_{\text{tors}}$ since $rs \in R \setminus \{0\}$.
- Let $x \in M_{\text{tors}}$, say $rx = 0$ for some $r \in R \setminus \{0\}$.
  $\forall t \in R, rtx = trx = t \cdot 0 = 0$.

Define $M_{\text{tf}} : M/M_{\text{tors}}$, called the torsion-free quotient / 无挠商 of M. Check that M_{tf} has no torsion elements $\neq 0$.

- Let $\overline{x} = x + M_{\text{tors}} \in M_{\text{tf}}, x \in M, r \in R/\{0\}, r\overline{x} = \overline{0}$, i.e. $rx \in M_{\text{tors}}$.
  $\Longrightarrow \exists s \in R \setminus \{0\}$ s.t. $\underbrace{sr}_{\in R \setminus \{0\}} x = 0 \Longrightarrow x \in M_{\text{tors}}$, i.e. $\overline{x} = \overline{0}$ in M_{tf}. $\square$

Observation $h \mid k \in R \Longrightarrow M[h] \subset M[k]$.

Now assume $t \in R \setminus \{0\}$ and R PID:

$$t \sim p_1^{a_1} \dots p_m^{a_m}, \quad p_i \text{ prime}, a \in \mathbb{Z}_{\geq 0}$$

Lemma $M[t] = \bigoplus_{i=1}^{m} M[p_i^{a_i}], M$ R-module.

Proof Suffices to show: $t = ab, a, b \in R$ coprime $\Longrightarrow M[t] = M[a] \oplus M[b]$.

Recall: $\exists u, v \in R$ s.t. $au + bv = 1$.
Let $x \in M[t], x = 1 \cdot x = \underbrace{aux}_{\in M[b]} + \underbrace{bvx}_{\in M[a]} \Longrightarrow M[t] = M[a] + M[b]$.
Let $x \in M[a] \cap M[b], x = uax + vbx = 0 + 0 = 0$ in M. $\square$

p prime in a PID R, M R-module, then $M[p] \subset M[p^2] \subset \dots$ Write $M[p^\infty] = \bigcup_{i \geq 1} M[p^i]$: submodule of M. So: If $\exists t \neq 0$ s.t. $M = M[t]$, then $M = \bigoplus_{p} M[p^\infty]$,

$\because M[p^n] = M[p^n] \cap M[t] = M[\gcd(p^n, t)] = M[p^a]$. If $p^a \parallel t$, then $M[p^\infty] = M[p^a]$.

Example R PID, $M = R/(t), p \in R$ prime.

- $p^a \mid t \implies (R/(t))[p^a] \simeq R/(p^a)$.
- $p^a \parallel t \implies (R/(t))[p^\infty] \simeq R/(p^a)$.

Proof

- $p^a \mid t, \overline{x} = x + (t) \in R/(t)$. $p^a\overline{x} = \overline{0} \iff \exists y, p^a x = ty$ in R. Put $s := t/p^a \in R$, then $x \in (s)$.
  So: $(R/(t))[p^a] = (s)/(t) \twoheadleftarrow R$ as R-modules via $sy + (t) \mapsfrom y$.
  Its kernel $= \{y \in R : t \mid sy\} = (p^a) \implies (R/(t))[p^a] \simeq R/(p^a)$.
- Note that $M := R/(t)$ satisfies $M = M[t]$. Hence $M[p^\infty] = M[p^a] \underset{(1)}{\simeq} R/(p^a)$. □

Second form of structure theorem for M finitely generated R-modules, R PID:

$$M \simeq \underbrace{R/(f_1) \oplus ... R/(f_k)}_{=M_{\text{tors}}=M[f_k]} \oplus E$$

where $f_1 \mid ... \mid f_k, f_i \in R \setminus \{0\}, \notin R^\times$.

$\forall p \in R$ prime, $p \mid f_k$, take $b_i(p)$ s.t. $p^{b_i(p)} \parallel f_i, \forall 1 \leq i \leq k$.

$$\begin{aligned} &\implies R/(f_i) = \bigoplus_{p \mid f_k} (R/(f_i))[p^\infty] \simeq \bigoplus_{p \mid f_k} (R/(f_i))\left[p^{b_i(p)}\right] \\ &\implies M_{\text{tors}} \simeq \bigoplus_{p \mid f_k} \bigoplus_{i} R/\left(p^{b_i(p)}\right) \end{aligned}$$

Note that invariant factors $f_1 \mid ... \mid f_k \longleftrightarrow$ elementary factors $b_1(p) \leq ... \leq b_k(p), \forall p$.

Now take $R = F[X]$, we obtain the 2nd form of rational canonical forms:

$A \in M_{n\times n}(F), n \in \mathbb{Z}_{\geq 1}, \mathrm{Min}_A = p_1^{a_1} ... p_m^{a_m}, p_i$ monic irred. in $F[X]$, distinct, $a_i \geq 1$. Then

$$A \underset{\text{conj}}{\sim} \mathrm{diag}(A_1, ..., A_m)$$

where $\forall 1 \leq j \leq m, A_j = \mathrm{diag}\left(C_{p_j^{b_{1,j}}}, ..., C_{p_j^{b_{r_j,j}}}\right), r_j \geq 1, 1 \leq b_{1,j} \leq ... \leq b_{r_j,j}$.

These data are unqiuely determined by A/conj, up to permutation of $p_1, ..., p_m$.

§ Existence part of structure theorem

R PID. The key tool is:

Lemma E free R-module, rank $= n \in \mathbb{Z}_{\geq 0}, N \subset E$ submodule. Then $\exists f_1, ..., f_n$ of E, $d_1, ..., d_n \in R$ s.t. $d_1 \mid ... \mid d_n$ and $d_1 f_1, ..., d_r f_r$ form a basis of N ($\implies N$ free)
where $r := \max\{i : d_i \neq 0\}$.

Proof of existence part in structure theorem

M finitely generated R-module, say generated by $x_1, ..., x_n$.
$E := R^{\oplus n} \twoheadrightarrow M$, its kernel $=: N \implies M \simeq E/N$. Take f_i, d_i, r from Lemma, then

$$\begin{aligned} M &\simeq \frac{Rf_1 \oplus ... \oplus Rf_n}{Rd_1 f_1 \oplus ... \oplus Rd_r f_r} \qquad \left(\frac{M_1 \oplus M_2}{N_1 \oplus N_2} \simeq \frac{M_1}{N_1} \oplus \frac{M_2}{N_2}\right) \\ &\simeq \left(\bigoplus_{i=1}^{r} R/(d_i)\right) \oplus R^{\oplus(n-r)}. \end{aligned}$$

□